

Foundations of Recurrent neural network

[Recurrent neural network]

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

1.0 Content Block

When the neural network has learned a certain percentage of the training data. When the minimum value of the mean-squared-error is satisfied. When the maximum number of training generations has been reached. The fitness function evaluates the stopping criterion as it receives the mean-squared error reciprocal from each network during training. Therefore, the goal of the genetic algorithm is to maximize the fitness function, reducing the mean-squared error. Other global (and/or evolutionary) optimization techniques may be used to seek a good set of weights, such as simulated annealing or particle swarm optimization.

2.0 Content Block

Long short-term memory (LSTM) is the most widely used RNN architecture. It was designed to solve the vanishing gradient problem. LSTM is normally augmented by recurrent gates called "forget gates". LSTM prevents backpropagated errors from vanishing or exploding. Instead, errors can flow backward through unlimited numbers of virtual layers unfolded in space. That is, LSTM can learn tasks that require memories of events that happened thousands or even millions of discrete time steps earlier. Problem-specific LSTM-like topologies can be evolved. LSTM works even given long delays between significant events and can handle signals that mix low and high-frequency components. Many applications use stacks of LSTMs, for which it is called "deep LSTM". LSTM can learn to recognize context-sensitive languages unlike previous models based on hidden Markov models (HMM) and similar concepts.

3.0 Content Block

Each weight encoded in the chromosome is assigned to the respective weight link of the network. The training set is presented to the network which propagates the input signals forward. The mean-squared error is returned to the fitness function. This function drives the genetic selection process. Many chromosomes make up the population; therefore, many different neural networks are evolved until a stopping criterion is satisfied. A common stopping scheme can be:

4.0 Content Block

An Elman network is a three-layer network (arranged horizontally as x, y, and z in the illustration) with the addition of a set of context units (u in the illustration). The middle (hidden) layer is connected to these context units fixed with a weight of one. At each time step, the input is fed forward and a learning rule is applied. The fixed back-connections save a copy of the previous values of the hidden units in the context units (since they propagate over the connections before the learning rule is applied). Thus the network can maintain a sort of state, allowing it to perform tasks such as sequence-prediction that are beyond the power of a standard multilayer perceptron. Jordan networks are similar to Elman networks. The context units are fed from the output layer instead of the hidden layer. The context units in a Jordan network are also called the state layer. They have a recurrent connection to themselves. Elman and Jordan networks are also known as "Simple recurrent networks" (SRN).

5.0 Content Block

$I_i(t)$: Input (if any) to node CTRNNs have been applied to evolutionary robotics where they have been used to address vision, co-operation, and minimal cognitive behaviour. Note that, by the Shannon sampling theorem, discrete-time recurrent neural networks can be viewed as continuous-time recurrent neural networks where the differential equations have transformed into equivalent difference equations. This transformation can be thought of as occurring after the post-synaptic node activation functions $y_i(t)$ have been low-pass filtered but prior to sampling. They are in fact recursive neural networks with a particular structure: that of a linear chain. Whereas recursive neural networks operate on any hierarchical structure, combining child representations into parent representations, recurrent neural networks operate on the linear progression of time, combining the previous time step and a hidden representation into the representation for the current time step. From a time-series perspective, RNNs can appear as nonlinear versions of finite impulse response and infinite impulse response filters and also as a nonlinear autoregressive exogenous model (NARX). RNN has infinite impulse response whereas convolutional neural network has finite impulse response. Both classes of networks exhibit temporal dynamic behavior. A finite impulse recurrent network is a directed acyclic graph that can be unrolled and replaced with a strictly feedforward neural network, while an infinite impulse recurrent network is a directed cyclic graph that cannot be unrolled. The effect of memory-based learning for the recognition of sequences can also be implemented by a more biological-based model which uses the silencing mechanism exhibited in neurons with a relatively high frequency spiking activity. Additional stored states and the storage under direct control by the network can be added to both infinite-impulse and finite-impulse networks. Another network or graph can also replace the storage if that incorporates time delays or has feedback loops. Such controlled states are referred to as gated states or gated memory and are part of long short-term memory networks (LSTMs) and gated recurrent units. This is also called Feedback Neural Network (FNN).

6.0 Content Block

Hierarchical recurrent neural network Hierarchical recurrent neural networks (HRNN) connect their neurons in various ways to decompose hierarchical behavior into useful subprograms. Such hierarchical structures of cognition are present in theories of memory presented by philosopher Henri Bergson, whose philosophical views have inspired hierarchical models. Hierarchical recurrent neural networks are useful in forecasting, helping to predict disaggregated inflation components of the consumer price index (CPI). The HRNN model leverages information from higher levels in the CPI hierarchy to enhance lower-level predictions. Evaluation of a substantial dataset from the US CPI-U index demonstrates the superior performance of the HRNN model compared to various established inflation prediction methods.

7.0 Content Block

Stacked RNN A stacked RNN, or deep RNN, is composed of multiple RNNs stacked one above the other. Abstractly, it is structured as follows Layer 1 has hidden vector h_1, t , parameters θ_1 , and maps $f_{\theta_1} : (x_0, t, h_1, t) \mapsto (x_1, t, h_1, t+1)$. Layer 2 has hidden vector h_2, t , parameters θ_2 , and maps $f_{\theta_2} : (x_1, t, h_2, t) \mapsto (x_2, t, h_2, t+1)$ Layer n has hidden vector h_n, t , parameters θ_n , and maps $f_{\theta_n} : (x_{n-1}, t, h_n, t) \mapsto (x_n, t, h_n, t+1)$. Each layer operates as a stand-alone RNN, and each layer's output sequence is used as the input sequence to the layer above. There is no conceptual limit to the depth of stacked RNN.